

AUTOMATED INCIDENTS RESPONSE SYSTEM USING SIEM AND SOAR (AIRS)

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With the increased volume of cyberattacks and the complexity of modern IT environments, traditional SOC's are not sufficient to find and respond to threats in a timely manner. This project aims to Design and Deploy an automated incident response system that uses security information and event management (SIEM) products and security orchestration, automation, and response (SOAR) tools to augment the capabilities of SOC's by automating detection and response procedures and reducing human error, and conducting incident analysis and responding faster.

Keywords (SIEM, SOAR, TheHive, Wazuh, incidents response, SOC, Attacks)

I. INTRODUCTION

The rapid evolution of cyber threats and the increasing complexity of modern IT infrastructures have significantly challenged the effectiveness of traditional Security Operations Centers (SOC's). Security Information and Event Management (SIEM) systems play a critical role in aggregating and analyzing security events; however, they often generate an overwhelming volume of alerts and lack automated response capabilities. Conversely, Security Orchestration, Automation, and Response (SOAR) platforms enable automated and coordinated incident handling, improving operational efficiency. Therefore, integrating SIEM and SOAR has emerged as a promising approach to enhance both detection and response processes.

Despite this potential, existing SIEM–SOAR integrations are frequently limited to theoretical models or partial implementations, with insufficient real-world validation. Furthermore, many solutions lack direct integration with enforcement mechanisms such as firewalls, resulting in delayed containment and continued reliance on manual intervention. These limitations highlight the need for a practical, fully integrated, and automated incident response framework.

This paper aims to design and implement an automated incident response system that integrates SIEM and SOAR technologies to improve detection accuracy and response speed. The proposed system is validated through simulated real-world attack scenarios, and its performance is evaluated using key metrics such as Mean Time to Detection (MTTD) and Mean Time to Response (MTTR).

The main contribution of this work is the development of a practical SIEM–SOAR integration framework that enables real-time automated response with direct network-level enforcement, such as blocking malicious IP addresses. Unlike prior studies, this work provides an applied implementation supported by quantitative performance evaluation, demonstrating significant improvements in incident response efficiency and consistency.

The remainder of this paper is organized as follows: Section II presents the background and related work. Section III describes the proposed system design and methodology. Section IV details the implementation and experimental results. Section V discusses performance evaluation. Finally, Section VI concludes the paper and outlines future work.

II. BACKGROUND AND LITERATURE REVIEW

A. Theoretical Background

SIEM systems provide centralized log collection, correlation, and real-time threat detection, but lack automated response capabilities. In contrast, SOAR platforms enable automation and orchestration of incident response through predefined playbooks. The integration of SIEM and SOAR combines accurate detection with automated response, improving efficiency and reducing response time in modern SOC's.

B. Related Work

Recent studies have explored SIEM–SOAR integration to enhance cybersecurity operations. Some works combined SIEM, SOAR, and AI to improve detection and response but lacked experimental validation. Other studies implemented Wazuh with TheHive, achieving near real-time alerting, yet without comprehensive performance evaluation. Additionally, several works provided theoretical comparisons or proposed architectures (e.g., SIEM–SOAR–honeypot), but were limited by complexity or lack of real-world validation.

C. Research Gap and Contribution

Existing research often lacks practical implementation, quantitative evaluation, and integration with enforcement mechanisms. This work addresses these gaps by proposing a fully implemented SIEM–SOAR system with automated response and direct network-level enforcement. The system is validated using simulated attacks and evaluated through performance metrics such as MTTD and MTTR, demonstrating improved response efficiency.

Comparison Matrix Between Literary Studies

Study Title	SIEM Used	SOAR Used	SIEM + SOAR Integration	AI Used	IDS/IP S	Firewall	Honeypot
Enhancing Cyber Resilience through the Convergence of SIEM, SOAR, and AI (2024)	✓	✓	✓	✓	✗	✗	✗
SIEM and Threat Intelligence: Protecting Applications with Wazuh and TheHive (2024)	✓	✓	✓	✗	✗	✗	✗
Mengenal SIEM dan SOAR – Pilar Utama Keamanan Informasi Modern (2025)	✓	✓	✓	✗	✗	✗	✗
Information Security Systems Design Using SIEM, SOAR, and Honeypot (2024)	✓	✓	✓	✗	✗	✗	✓
AIRS System (Preliminary)	✓	✓	✓	✗	✗	✓	✗

III. METHODOLOGY (ANALYSIS AND MODELING)

This study adopts a hybrid methodology combining Agile and DevOps practices to design, implement, and evaluate an automated incident response system integrating SIEM and SOAR technologies. The approach enables iterative development, continuous testing, and incremental improvement of detection rules and response workflows.

From an analysis perspective, the current SIEM-based system is examined to identify its capabilities and limitations, particularly its reliance on manual intervention and inability to perform automated response actions. Based on this analysis, a proposed model is developed by integrating SIEM with SOAR to enable automated detection, analysis, and response within a unified architecture. The system model consists of three main layers: (1) data collection and log analysis using SIEM (Wazuh), (2) alert management and case handling using SOAR tools (TheHive), and (3) automation and orchestration through Shuffle playbooks. Alerts generated by SIEM are forwarded to the SOAR platform, where automated workflows are triggered to enrich, analyze, and respond to incidents.

To validate the proposed model, simulated attack scenarios such as brute-force and malware attacks are executed. The system performance is evaluated using key performance indicators, including Mean Time to Detection (MTTD) and Mean Time to Response (MTTR), to measure improvements in detection speed, response efficiency, and overall system effectiveness.

IV. SYSTEM DESIGN AND IMPLEMENTATION

The proposed system is designed as an integrated automated incident response framework that combines SIEM and SOAR technologies to enhance cybersecurity operations. The architecture follows a layered design approach, where data collection, analysis, automation, and incident handling are logically separated to ensure scalability and efficiency.

At the data collection layer, endpoint devices such as Windows and Linux systems are equipped with Wazuh agents that continuously monitor and collect security-related logs, including authentication events, system activities, and potential anomalies. These logs are securely forwarded to the Wazuh manager, where they undergo parsing, correlation, and rule-based analysis to identify suspicious behaviors and assign severity levels to detected events.

Once high-severity alerts are generated, they are transmitted to the SOAR layer through a webhook integration. The Shuffle platform acts as the orchestration engine, where automated workflows are executed to process alerts, extract relevant information, and apply predefined logic for decision-making. This enables the system to filter noise, prioritize incidents, and initiate appropriate response actions without manual intervention.

For incident management, the system integrates with TheHive platform, which is responsible for creating, tracking, and managing security cases. Each alert is transformed into a structured incident, allowing security analysts to investigate, document, and coordinate response activities efficiently. Additionally, automated response mechanisms are implemented, such as blocking malicious IP addresses, terminating suspicious processes, and sending real-time notifications to analysts, ensuring rapid containment of threats. The implementation phase involved deploying Wazuh SIEM on an Ubuntu-based environment, including its manager, indexer, and dashboard components, followed by the installation and configuration of agents on monitored endpoints. The SOAR platform (Shuffle) was deployed using Docker containers to ensure flexibility and ease of integration, while TheHive was configured as the central incident response platform.

Overall, the integration of SIEM and SOAR technologies provides a comprehensive solution that enables real-time threat detection, automated decision-making, and efficient incident response. The system reduces manual effort, minimizes response time, and improves the overall effectiveness of security operations in modern IT environments.

V. RESULTS AND KPIS

The implemented system was evaluated through a series of controlled experiments designed to validate its effectiveness in detecting, analyzing, and responding to cybersecurity incidents. The evaluation focused on measuring the performance of the integrated SIEM–SOAR framework in terms of detection accuracy, response time, and automation efficiency under simulated attack scenarios.

To assess the system, a brute-force attack scenario was conducted by generating multiple failed logins attempts from different IP addresses within a short time frame. The Wazuh SIEM successfully detected these events using predefined correlation rules and escalated them into high-severity alerts. The system demonstrated accurate event correlation by identifying repeated authentication failures as a potential brute-force attack rather than isolated incidents.

Once the alerts were generated, they were forwarded to the Shuffle SOAR platform, where automated workflows were triggered. The system executed predefined response actions, including blocking malicious IP addresses and generating incident cases in TheHive. This automated pipeline significantly reduced the need for manual intervention and ensured consistent handling of security incidents.

detected incidents were correctly classified without misidentification. Additionally, the False Positive Rate was recorded at 0%, demonstrating the effectiveness of the rule-based detection and filtering mechanisms in minimizing alert noise. The Specificity reached 100%, further validating the system’s ability to correctly distinguish between malicious and benign activities.

Furthermore, the integration of SIEM and SOAR technologies contributed to a significant improvement in response time. Automated workflows enabled immediate execution of mitigation actions upon detection, reducing the delay between detection and response compared to traditional manual processes. The system also enhanced operational efficiency by centralizing incident management and providing real-time notifications to security analysts.

Overall, the experimental results demonstrate that the proposed system provides a highly effective and reliable solution for automated incident detection and response. The achieved KPIs highlight the system's capability to deliver accurate threat detection, minimize false alerts, and ensure rapid and consistent response to cybersecurity incidents in real-world environments.

VI. CONCLUSION AND FUTURE WORK

This paper presented an integrated automated incident response framework that combines SIEM with SOAR technologies to address the growing complexity of cybersecurity operations. Unlike traditional approaches that rely heavily on manual intervention, the proposed system introduces a tightly coupled architecture that enables real-time threat detection, automated decision-making, and orchestrated response execution within a unified environment.

The main research contribution of this work lies in the practical integration of Wazuh SIEM, Shuffle SOAR, and TheHive into a cohesive and fully automated pipeline. This integration demonstrates how heterogeneous security tools can be orchestrated to achieve end-to-end incident handling, from log collection and correlation to automated mitigation and case management.

Furthermore, the system provides empirical evidence that rule-based detection combined with automated workflows can significantly reduce response time while maintaining high detection accuracy.

The experimental evaluation confirmed that the system achieves high performance across key metrics, including perfect accuracy and precision with zero false positives under the tested scenarios. While these results highlight the effectiveness of the proposed approach, they also reflect the controlled nature of the experimental setup. Therefore, a critical interpretation suggests that system performance may vary under more complex and dynamic real-world environments, where attack patterns are less predictable and noise levels are significantly higher.

Limitations:

Despite its strengths, the proposed system has several limitations that must be acknowledged. First, the evaluation was conducted using a limited set of attack scenarios, primarily focusing on brute-force attacks and predefined rule-based detections, which may not fully represent the diversity of modern cyber threats. Second, the reliance on static detection rules in Wazuh may limit the system's ability to detect novel or zero-day attacks. Third, scalability was not extensively tested under high-volume log ingestion or large-scale enterprise environments, which may impact system performance and response latency. Additionally, the automation logic implemented in the SOAR platform depends on predefined workflows, which may require continuous tuning and updating to remain effective against evolving threats.

In terms of future work, several enhancements can be pursued to address these limitations and extend the system's capabilities. Expanding the detection scope to include advanced attack vectors such as ransomware, lateral movement, and data exfiltration would improve the robustness of the system. Integrating threat intelligence feeds can further enrich alert context and support more informed automated decisions. Moreover, incorporating machine learning techniques for anomaly detection and adaptive response can enhance the system's ability to identify unknown threats and dynamically adjust response strategies.

Finally, future research should focus on large-scale performance evaluation and real-world deployment to assess the system's scalability, reliability, and operational impact in enterprise environments. Enhancing automation with context-aware and risk-based decision-making mechanisms will also be essential to move toward fully autonomous security operations centers (SOCs).